

Asynchronous Multiuser Reception for OFDM in Underwater Acoustic Communications

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Abstract—Recently significant progress has been made on point-to-point underwater acoustic communications, and the interest has grown on the application of those techniques in multiuser communication settings, where the asynchronous nature of multiuser communication poses a grand challenge. This paper develops a time-asynchronous multiuser reception approach for orthogonal frequency-division multiplexing (OFDM) transmissions in underwater acoustic channels. The received data burst is segmented and apportioned to multiple processing units in an overlapped fashion, where the length of the processing unit depends on the maximum asynchronism among users on the OFDM block level. Interference cancellation is adopted to reduce the interblock interference between overlapped processing units. Within each processing unit, the residual inter-block interference from multiple users is aggregated as one external interference which can be parameterized. Multiuser channel estimation, data detection, and interference mitigation are then carried out in an iterative fashion. Simulation and emulated experimental results demonstrate the robustness of the proposed receiver with signal asynchronism among multiple users in both time-invariant and time-varying environments. It is observed that the receiver decoding performance degrades as the channel time variation and the maximum relative delay among users increase.

Index Terms—Asynchronous multiuser communications, OFDM, underwater acoustic channels, overlapped truncation, interference aggregation.

I. INTRODUCTION

UNDERWATER acoustic (UWA) communications and networking have been under extensive investigation in recent years. On the physical layer, significant advances have been witnessed for both single-carrier and multicarrier transmissions; see, e.g. [2]–[17], and references therein. Interest is rapidly growing on the integration of those latest point-to-point communication techniques to practical underwater networked systems with multiple users.

This paper considers the application of OFDM modulation to multiuser underwater acoustic communications. Should the signals from multiple users be synchronized at the receiver on the OFDM block level, multiuser reception can be viewed as a distributed multi-input multi-output (MIMO) problem, as

considered in [18]–[20]. However, due to large propagation delays and lack of a well-defined network infrastructure, synchronization is an extremely challenging task in the distributed underwater acoustic system.

A. Asynchronous OFDM Research in Radio Channels

Asynchronous multiuser reception has been extensively investigated with code-division multiple access (CDMA) transmissions; see e.g., [21]–[23] and references therein. The impact of the asynchronism of users on orthogonal frequency-division multiple access (OFDMA) uplink transmission has been investigated in [24], [25]. Related work on the asynchronous OFDM receiver is limited [24]–[30], and can be broadly grouped into two main categories.

The works in the first category focus on the demodulation and decoding modules of the receiver, assuming that perfect channel knowledge is available. In [26], a maximum *a posteriori* (MAP) receiver over a three-dimensional trellis was developed for asynchronous multiuser OFDMA transmissions, where the trellis is expanded by the user index, time index and subcarrier index. This method has very high computational complexity when the number of subcarriers is large. Using the same truncation method as in the synchronous multiuser reception to divide the received signal into individual processing units, Ref. [27] pointed out that the conventional frequency-domain approaches for decoding and interference suppression are not effective, as the orthogonality of subcarriers of the misaligned users is lost due to the misalignment of the otherwise-orthogonal users' signals. Hence, the spatial-time filtering techniques were employed in [27]–[29] to exploit the spatial diversity of multiple receivers. In [31], a general message-passing based equalization method was proposed in the presence of interference.

The works in the second category focus on channel estimation in asynchronous OFDM systems. A subspace based semi-blind channel estimator was proposed in [30] to separate the desired user's channel vector from channel vectors of other users. The method, although often very useful, can be expected to suffer when the channel has fast time-variation.

Apparently, a complete receiver design for asynchronous OFDM with both channel estimation and equalization is not available even for radio channels. In addition, methods developed for radio channels may not be directly applicable for fast-varying underwater acoustic channels.

B. This Work

Most works for underwater acoustic OFDM are focused on point-to-point communications, e.g., [2], [3], [6], and

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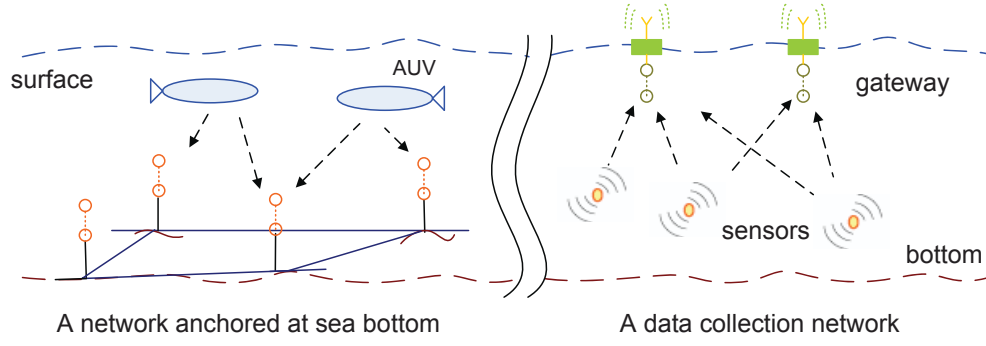


Fig. 1. Two example underwater networks. The nodes anchored at sea bottom in the first network are connected to a control center via cables. The gateways in the second network can communicate with satellites or ships using radio waves.

extensions to cooperative OFDM usually assume block-level synchronism among users [18]. In this paper, we propose a multiuser receiver for the asynchronous OFDM transmissions in underwater acoustic channels. The main contributions are the following.

- To our best knowledge, this is the first work on the asynchronous multiuser OFDM reception in underwater acoustic channels. As the proposed approach allows the simultaneous transmission of multiple users without performing coordination, it has practical impacts on the MAC protocol design for certain emerging underwater acoustic networks [32]–[35].
- To achieve the asynchronous multiuser reception, we develop a burst-by-burst processing method. In particular, an *over-lapped truncation approach* is proposed to retain the desired block-information from all the users within one truncation. By taking the aggregated cochannel interference caused by the asynchronism as one interference, the asynchronous multiuser problem is converted to a quasi-synchronous multiuser problem with interference contamination. Moreover, the overlapped truncation and the interference aggregation enable the interference parameterization and cancellation regardless of the number of users, hence simplifying the receiver design.
- Extensive simulation and emulated experimental results are used to validate the performance of the proposed receiver in both time-invariant and time-varying scenarios. These results highlight the dependence of the decoding performance on the maximum relative delay among users.

Although the network throughput can be greatly increased by allowing simultaneous transmissions from asynchronous multiusers, the computational complexity of the asynchronous multiuser reception is much higher than its synchronous counterpart, hence requiring a high computational capability of the processing unit. Two example networks are shown in Fig. 1, where in the left one multiple autonomous underwater vehicles (AUVs) communicate to a fixed cabled network on the sea bottom [36], [37], and in the right one multiple sensor nodes communicate data back to the buoys on the water surface. In both cases, the receiving nodes, either anchored at the sea bottom or attached to a surface buoy, are assumed to have the processing power needed to decode simultaneous transmissions from multiple users.

Remark 1: In this paper, we mainly focus on the scenario that all the users are stationary or have similar moving speeds. Investigation on the asynchronous multiuser reception in a scenario where different users have different moving speeds is beyond the scope of this work. By assuming the *synchronous* transmission from multiple user, receiver designs dealing with quite different speeds of different users have recently been studied in [18]–[20].

The rest of the paper is organized as follows. A burst-by-burst asynchronous multiuser OFDM reception approach is developed in Section II, with several key modules detailed in Section III. Investigation on the time duration of the aggregated interference in an example network is provided in Section IV. Simulations and emulated experimental results are presented in Sections V and VI, respectively. Conclusions are contained in Section VII.

Notation: Bold upper case letters and lower case letters are used to denote matrices and column vectors, respectively. $(\cdot)^T$ and $(\cdot)^H$ denote transpose and Hermitian transpose, respectively. $\propto$ denotes equality of functions up to a scaling factor. $[\mathbf{a}]_m$ denotes the m th element of vector $\mathbf{a}$, and $[\mathbf{A}]_{m,k}$ denotes the (m, k) th element of matrix $\mathbf{A}$.

II. THE PROPOSED RECEIVER FOR ASYNCHRONOUS OFDM TRANSMISSIONS

Consider an underwater system consisting of N_u asynchronous users and a receiver equipped with N_r receiving hydrophones, and $N_r > N_u$. Assume that all users use the zero-padded (ZP) OFDM block transmission with an identical parameter set, which includes the center frequency f_c , bandwidth B , number of subcarriers K , symbol duration T and a silent guard interval of length T_g with $T_{bl} = T + T_g$. The k th frequency subcarrier is denoted as

$$f_k = f_c + \frac{k}{T}, \quad k = -\frac{K}{2}, \dots, \frac{K}{2} - 1. \quad (1)$$

Out of the total K subcarriers, there are data subcarriers for information delivery, pilot and null subcarriers for channel and noise variance estimation. Define $s_\mu[k; n]$ as the symbol transmitted at the k th subcarrier of the n th block from the μ th user, and assume N_{bl} blocks in each transmission burst. The transmitted signal from the μ th user is

$$\tilde{x}_\mu(t) = \sum_{n=1}^{N_{bl}} \tilde{s}_\mu(t - nT_{bl}; n), \quad t \in [0, N_{bl}T_{bl}] \quad (2)$$

where the n th transmitted block is expressed as

$$\tilde{s}_\mu(t; n) = 2\text{Re} \left(\sum_{k=-K/2}^{K/2-1} s_\mu[k; n] e^{j2\pi f_k t} g(t) \right), \quad t \in [0, T_{\text{bl}}] \quad (3)$$

with $g(t)$ being a rectangular window which has nonzero support within $[0, T]$.

Using a path-based channel model [10], the channel impulse response between the μ th user and the ν th receiving hydrophone during the transmission of the n th block is expressed as

$$h_{\nu,\mu}(\tau, t; n) = \sum_{p=1}^{N_{\text{pa},\nu,\mu}^{(n)}} A_{\nu,\mu,p}^{(n)} \delta(\tau - (\tau_{\nu,\mu,p}^{(n)} - a_{\nu,\mu,p}^{(n)} t)), \quad (4)$$

where $N_{\text{pa},\nu,\mu}^{(n)}$ is the number of discrete channel paths, $A_{\nu,\mu,p}^{(n)}$, $\tau_{\nu,\mu,p}^{(n)}$ and $a_{\nu,\mu,p}^{(n)}$ are the amplitude, initial delay, and Doppler scaling factor of the p th path, respectively. The Doppler scaling factor $a_{\nu,\mu,p}^{(n)}$ is related to the speed of the p th path $v_{\nu,\mu,p}^{(n)}$ via $a_{\nu,\mu,p}^{(n)} = v_{\nu,\mu,p}^{(n)}/c$, where c is the sound speed in water. The triplets $(A_{\nu,\mu,p}^{(n)}, \tau_{\nu,\mu,p}^{(n)}, a_{\nu,\mu,p}^{(n)})$ are assumed independent across blocks. The signal from the μ th user at the ν th receiving hydrophone can be expressed as

$$\tilde{y}_{\nu,\mu}(t) = \sum_{n=1}^{N_{\text{bl}}} \tilde{y}_{\nu,\mu}(t - nT_{\text{bl}}; n), \quad t \in [0, N_{\text{bl}}T_{\text{bl}}] \quad (5)$$

with

$$\tilde{y}_{\nu,\mu}(t; n) = \sum_{p=1}^{N_{\text{pa},\nu,\mu}^{(n)}} A_{\nu,\mu,p}^{(n)} \tilde{s}_\mu((1 + a_{\nu,\mu,p}^{(n)})t - \tau_{\nu,\mu,p}^{(n)}). \quad (6)$$

Let ε_μ denote the time-of-arrival the μ th user, which can be obtained at the receiver by detecting the preamble of this user. On the block level, we can assume that $\varepsilon_\mu \leq T_{\text{bl}}/2$, as integer block delays can be incorporated by reindexing the blocks. Without loss of generality, we assume that $0 = \varepsilon_1 \leq \varepsilon_2 \leq \dots \leq \varepsilon_{N_u} \leq T_{\text{bl}}/2$, and define $\varepsilon_{\text{max}} := \varepsilon_{N_u}$. The passband signal at the ν th receiving element is the superposition of N_u waveforms,

$$\tilde{y}_\nu(t) = \sum_{\mu=1}^{N_u} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu) + \tilde{w}_\nu(t), \quad (7)$$

where $\tilde{y}_{\nu,\mu}(t)$ is the signal at the ν th receiving element from the μ th user, and $\tilde{w}_\nu(t)$ denotes the ambient noise.

A. Overlapped Truncation

To facilitate the decoding operation at the receiver, the received signal is usually truncated into individual processing units. For synchronous multiuser transmissions, the truncation can be easily carried out according to the block structure of the transmitted signal. However, for asynchronous transmissions, the block structure of the transmitted signal at the receiver is destroyed. As shown in Fig. 2, a block from one user can collide with multiple blocks from other users. Different truncation methods could lead to different decoding schemes [25]. One existing method is to synchronize the truncation to the time-of-arrival of one desired user [27]–[30], with each truncation having a block length T_{bl} , including the complete

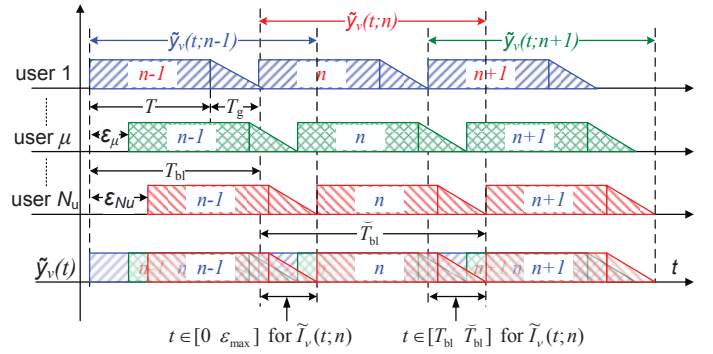


Fig. 2. Illustration of the overlapped partition of the received signal and the aggregated interference in an asynchronous N_u -user system.

information of one block from the desired user and partial block information from others. However, this method is not effective when the overlap length of the desired user and others is large.

In this work, we consider an overlapped truncation method as shown in Fig. 2, where each truncated block length is $\tilde{T}_{\text{bl}} := T_{\text{bl}} + \varepsilon_{\text{max}}$. The n th truncated block consists of the information from $(3N_u - 2)$ transmitted blocks, including i) part of the $(n - 1)$ th blocks from users $2 \sim N_u$ at the beginning of this truncation, ii) complete information of the n th blocks from the N_u users, and iii) part of the $(n + 1)$ th blocks from users $1 \sim (N_u - 1)$ at the end of this truncation,

$$\begin{aligned} \tilde{y}_\nu(t; n) = & \underbrace{\sum_{\mu=1}^{N_u} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu; n)}_{\text{desired OFDM signal}} + \underbrace{\sum_{\mu=2}^{N_u} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu + T_{\text{bl}}; n - 1) + \sum_{\mu=1}^{N_u-1} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu - T_{\text{bl}}; n + 1)}_{\text{interference from the preceding and succeeding processing units}}, \end{aligned} \quad (8)$$

for $t \in [0, \tilde{T}_{\text{bl}}]$.

With (8), an optimal receiver can be designed by treating the asynchronous N_u -user problem as a synchronous $(3N_u - 2)$ -user problem [21]. However, solving the problem therein usually requires more efforts than solving a typical synchronous $(3N_u - 2)$ -user problem, since the orthogonality of subcarriers of the $(2N_u - 2)$ misaligned users in (8) is destroyed.

B. Interference Aggregation

In this paper, rather than modeling the partial block interferences from $(2N_u - 2)$ users individually with the transmitted signal, we use an *interference aggregation* concept to treat the aggregated IBI as one interfering waveform as shown in Fig. 2, which is formulated as

$$\tilde{I}_\nu(t; n) = \begin{cases} \sum_{\mu=2}^{N_u} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu + T_{\text{bl}}; n - 1), & t \in [0, \varepsilon_{\text{max}}] \\ 0, & t \in [\varepsilon_{\text{max}}, T_{\text{bl}}] \\ \sum_{\mu=1}^{N_u-1} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu - T_{\text{bl}}; n + 1), & t \in [T_{\text{bl}}, \tilde{T}_{\text{bl}}]. \end{cases} \quad (9)$$

The bandwidth of the aggregated interference is taken as identical to that of the useful signal denoted by B . Given

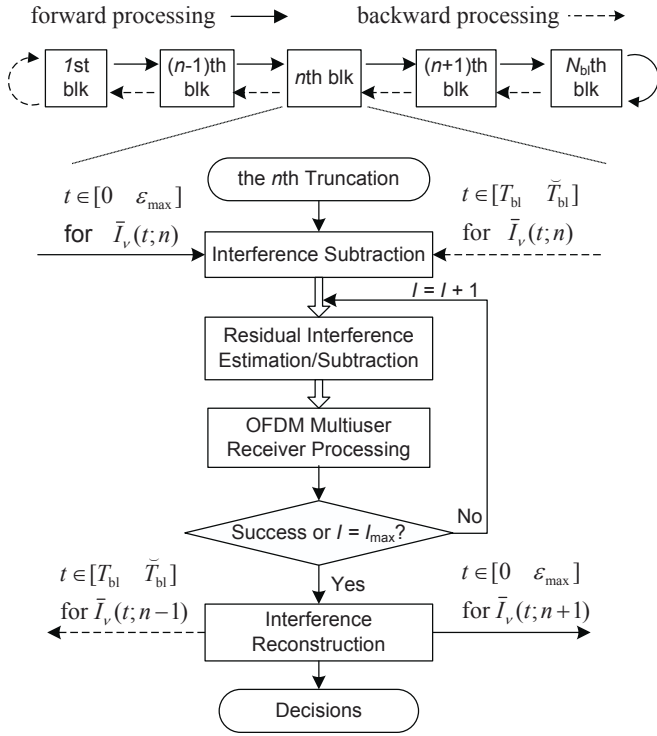


Fig. 3. Illustration of the proposed burst-by-burst asynchronous multiuser receiver with iterative forward/backward processing, with N_{bl} blocks in each burst.

the maximum delay $\varepsilon_{\max}$, one can see that the number of the degrees-of-freedom (DoF) of the interfering waveform, i.e., the time-bandwidth product of the aggregated interference $[2B\varepsilon_{\max}]$, does not change as the number of asynchronous users increases.

We then have (8) reformulated as

$$\tilde{y}_\nu(t; n) = \sum_{\mu=1}^{N_u} \tilde{y}_{\nu,\mu}(t - \varepsilon_\mu; n) + \tilde{I}_\nu(t; n) + \tilde{w}_\nu(t; n) \quad (10)$$

for $t \in [0, \tilde{T}_{bl}]$. Hence, the asynchronous N_u -user problem can be regarded as a synchronous N_u -user problem in the presence of an external interference.

C. The Overall Structure of the Proposed Receiver

As shown in [26], the IBI arising from the asynchronism of users can be optimally mitigated by performing a joint decoding of blocks in one transmission from all users. However, this is usually computationally prohibitive. Leveraging the overlapped truncation method and the interference aggregation concept, we next propose a multiuser reception approach by performing a burst-by-burst decoding with interference cancellation.

Different from the external interference considered in [38], the time-domain input-output relationship in (8) shows the interference term in (10) actually consists of part of useful signals corresponding to the $(n-1)$ th and the $(n+1)$ th transmitted blocks from the N_u users. If these blocks have been successfully decoded or estimates of transmitted symbols within these blocks are available, one can get initial estimates of the interferences that spill over from these blocks to the n th

block, and thus take the estimates as the *a priori* knowledge of the interferences. After subtracting initial estimates of the interferences from the received signal, the joint multiuser decoding and cancellation of the aggregated residual interference can be performed in the n th block.

The proposed asynchronous multiuser receiver for each block consists of the following four steps¹.

- *Step (1) Interference subtraction:* With the estimated interference passed from the preceding and the succeeding processing units, interference subtraction can be carried out prior to the multiuser decoding;
- *Step (2) Joint multiuser processing with residual interference cancellation:* Techniques for synchronous multiuser decoding and approaches for external interference cancellation can be used;
- *Step (3) Interference reconstruction:* Based on the multiuser decoding results, the interference of the current block to the preceding and the succeeding blocks will be reconstructed.
- *Step (4) Iteration among Steps (1) ~ (3):* To improve the interference cancellation performance, an iterative forward/backward processing of blocks within one burst can be performed, as shown in Fig. 3. As the iteration goes on, the accuracy of interference estimation improves gradually, thus leading to a better multiuser decoding performance, which in turn boosts the performance of interference estimation.

A detailed description of three modules in the proposed receiver will be provided in Section III.

D. Discussions of the Proposed Receiver

Compared with a synchronous multiuser receiver, the proposed asynchronous multiuser receiver can only perform multiuser decoding after receiving the whole burst from all users, hence suffering a processing latency. Similar to the Viterbi algorithm for channel equalization, the latency can be reduced by using the *sliding block* techniques proposed in, e.g. [39], [40], which leads to a batch-by-batch processing, with the batch size depending on the maximum tolerant latency and the storage capability of the receiver.

When the batch size is one, the proposed receiver degrades to a *block-by-block* receiver. Relative to the block-by-block decoding, burst-by-burst decoding leverages the decoding results of preceding and succeeding blocks for interference cancellation, hence enjoys a better decoding performance; however, it suffers a processing latency of the batch size.

Hereafter, we mainly focus on the receiver design for burst-by-burst processing, and the block-by-block receiver can be obtained as a special case whose performance will also be tested in Sections V and VI.

Remark 2: In a practical OFDM system with N_u users, the time-of-arrival of the signal from each user can be estimated by transmitting a preamble prior to the useful data stream at the transmitter side. Preambles for N_u users could be obtained from a set of m-sequence waveforms or a set of Costas

¹The proposed scheme can be extended to both cyclic-prefixed (CP)-OFDM transmissions and single-carrier block transmissions if the OFDM multiuser decoding module is replaced by a similar module tailored to the corresponding signalling format.

waveforms, which have low cross correlation values. At the receiver side, N_u matched filters are used to detect the arrival of the useful signal from each user in the presence of multiuser interference. Note that the matched filter gain is equal to the time-bandwidth product of the preamble waveform. A large time duration or bandwidth is desirable when the multiuser interference is severe.

III. DESCRIPTIONS OF RECEIVER MODULES

This section provides detailed descriptions on different modules of the proposed receiver for asynchronous OFDM transmissions. The modules are presented according to their order in the receiver processing.

Prior to the interference subtraction, the passband-to-baseband downshifting and baseband lowpass filtering are performed. Define $y_\nu(t; n)$ as the baseband waveform corresponding to the passband signal $\tilde{y}_\nu(t; n)$. In the sequel, we focus on the receiver modules operating in the baseband.

A. Interference Subtraction

Assume the *a priori* knowledge of the time-domain interference waveform in baseband which is denoted by $\tilde{I}_\nu(t; n)$ (its availability will be discussed in Section III-D). Prior to the receiver processing, subtraction of the initial interference estimate from $y_\nu(t; n)$ in (10) leads to²

$$\bar{y}_\nu(t; n) = \sum_{\mu=1}^{N_u} y_{\nu,\mu}(t - \varepsilon_\mu; n) + \eta_\nu(t; n) + w_\nu(t; n) \quad (11)$$

for $t \in [0, \tilde{T}_{bl}]$, where $\eta_\nu(t; n)$ denotes the residual interference. As the residual interference has an identical time duration and bandwidth as the interference $I_\nu(t; n)$, the number of DoF of $\eta_\nu(t; n)$ is also $\lceil 2B\varepsilon_{\max} \rceil$.

In the initial forward *block-to-block* decoding of the proposed receiver, the *a priori* knowledge of the interference from the subsequent block is not available. Hence, we set it as zero at the beginning. Once all the blocks have been processed, the initial estimates of the interference spilled over from both neighboring blocks are available. During the following processing, the latest estimates of the decoded blocks are used for interference cancellation.

B. Frequency-Domain Oversampling

Define α as the frequency-domain oversampling factor. The m th frequency component of the time-domain signal $\bar{y}_\nu(t; n)$ can be obtained via

$$z_\nu[m; n] = \int_0^{\tilde{T}_{bl}} \bar{y}_\nu(t; n) e^{-j2\pi \frac{m}{\alpha T} t} dt, \quad (12)$$

for $m = -\alpha K/2, \dots, \alpha K/2 - 1$. Similarly, the m th frequency component $\eta_\nu[m; n]$ of the residual interference $\eta_\nu(t; n)$ can be obtained.

Stack the frequency components $z_\nu[m; n]$ and $\eta_\nu[m; n]$ into vectors $\mathbf{z}_\nu[n]$ and $\boldsymbol{\eta}_\nu[n]$ of size $\alpha K \times 1$, respectively. The

input-output relationship in the frequency domain corresponding to (11) can be compactly expressed as

$$\mathbf{z}_\nu[n] = \sum_{\mu=1}^{N_u} \mathbf{\Lambda}(\varepsilon_\mu) \mathbf{H}_{\nu,\mu}[n] \mathbf{s}_\mu[n] + \boldsymbol{\eta}_\nu[n] + \mathbf{w}_\nu[n], \quad (13)$$

where $\mathbf{w}_\nu[n]$ is the ambient noise vector, and the first summation term denotes the desired OFDM signal, with $\mathbf{s}_\mu[n]$ being a $K \times 1$ vector formed by the transmitted symbols at all subcarriers from the μ th user, $\mathbf{H}_{\nu,\mu}[n]$ denoting an $\alpha K \times K$ channel matrix between the μ th user and the ν th receiving element, and $\mathbf{\Lambda}(\varepsilon_\mu)$ instantiating a generic diagonal matrix defined as

$$[\mathbf{\Lambda}(\tau)]_{m,m} = e^{-j2\pi \frac{m}{\alpha T} \tau}. \quad (14)$$

Remark 3: Using the baseband sampling rate $B = K/T$, there are $\lceil \tilde{T}_{bl} B \rceil = \lceil \tilde{T}_{bl} K/T \rceil$ samples in each processing unit. To avoid information loss during the Fourier transform, the frequency-domain oversampling factor α should satisfy $\alpha K \geq \tilde{T}_{bl} K/T$, i.e., $\alpha T \geq \tilde{T}_{bl}$. Based on $\varepsilon_{\max} < T_{bl}/2$, we require $\alpha \geq 3(1 + T_g/T)/2$. Taking $\alpha = 2$ for example, the guard interval should satisfy $T_g \leq T/3$.

C. Multiuser Channel Estimation and Data Decoding with Interference Cancellation

Based on the frequency-domain input-output relationship in (13), there are several unknowns to estimate, including: i) the channel matrices $\{\mathbf{H}_{\nu,\mu}[n]\}$, ii) the information symbols in $\{\mathbf{s}_\mu[n]\}$, and the interference $\{\boldsymbol{\eta}_\nu[n]\}$. To this end, we propose a receiver structure for iterative multiuser channel estimation and data decoding and residual interference estimation within each block, as shown in Fig. 3.

During the iterative processing, the multiuser decoding step takes the residual interference estimate in the last iteration as the input, and based on the estimated desired OFDM component after subtracting residual interference from $\mathbf{z}_\nu[n]$, it outputs the estimated channel matrices and information symbols. Taking the output of multiuser decoding step as input, the residual interference cancellation step will subtract the reconstructed OFDM component from the frequency measurement $\mathbf{z}_\nu[n]$, so as to update the residual interference estimate which is then fed back for the next iteration. Once the parity check conditions of the channel decoders of all users are satisfied, or the number of iterations reaches a predetermined threshold $I_{\max}$, the iteration stops.

For the initialization of the iterative receiver, we first treat the residual interference as the ambient noise to get the initial estimates of channel matrices and information symbols, which are then used to initialize the iterative operation.

The two critical components of the receiver are described next.

1) *Parameterized Residual Interference Estimation and Subtraction:* With the estimated channel matrices $\{\hat{\mathbf{H}}_{\nu,\mu}[n]\}$ and transmitted symbols $\{\hat{\mathbf{s}}_\mu[n]\}$ from initialization or from the last iteration, the frequency-domain measurements after subtracting the OFDM components are

$$\hat{\boldsymbol{\eta}}_\nu[n] = \mathbf{z}_\nu[n] - \sum_{\mu=1}^{N_u} \mathbf{\Lambda}(\varepsilon_\mu) \hat{\mathbf{H}}_{\nu,\mu}[n] \hat{\mathbf{s}}_\mu[n], \quad (15)$$

²For receiver implementation, the interference subtraction can be performed with the baseband discrete samples.

which includes both the residual interference, and the equivalent ambient noise consisting of both ambient noise, and the channel and information symbol estimation errors.

To facilitate the residual interference estimation, we introduce a parameterized interference representation. Based on the time-bandwidth product $N_I := \lceil B\varepsilon_{\max} \rceil$, the interference in baseband can be approximated by the Fourier series expansion

$$\eta_\nu(t; n) \approx \begin{cases} \sum_{l=-N_I/2}^{N_I/2} c_{l,\nu,\text{he}} e^{j2\pi \frac{l}{\varepsilon_{\max}} t}, & t \in [0, \varepsilon_{\max}] \\ 0, & t \in [\varepsilon_{\max}, T_{\text{bl}}] \\ \sum_{l=-N_I/2}^{N_I/2} c_{l,\nu,\text{ta}} e^{j2\pi \frac{l}{\varepsilon_{\max}} t}, & t \in [T_{\text{bl}}, T_{\text{bl}} + \varepsilon_{\max}] \end{cases} \quad (16)$$

where $\{c_{l,\nu,\text{he}}\}$ and $\{c_{l,\nu,\text{ta}}\}$ represent the Fourier series coefficients of the front and end portions of the interference, respectively.

Following the time-to-frequency transform in (12), the m th frequency sample of $\eta_\nu(t; n)$ is formulated as

$$\eta_\nu[m; n] = (u_{l,\nu,\text{he}} + u_{l,\nu,\text{ta}} e^{-j2\pi \frac{m}{\alpha T} T_{\text{bl}}}) \Gamma[m, l], \quad (17)$$

where

$$u_{l,\nu,\text{he}} := \frac{\varepsilon_{\max}}{T} c_{l,\nu,\text{he}}, \quad u_{l,\nu,\text{ta}} := \frac{\varepsilon_{\max}}{T} c_{l,\nu,\text{ta}},$$

$$\Gamma[m, l] := \frac{\sin\left(\pi \left(\frac{m}{\alpha T} - \frac{l}{\varepsilon_{\max}}\right) \varepsilon_{\max}\right)}{\pi \left(\frac{m}{\alpha T} - \frac{l}{\varepsilon_{\max}}\right) \varepsilon_{\max}} e^{-\pi \left(\frac{m}{\alpha T} - \frac{l}{\varepsilon_{\max}}\right) \varepsilon_{\max}}.$$

Stack $u_{l,\nu,\text{he}}$ and $u_{l,\nu,\text{ta}}$ into vectors $\mathbf{u}_{\nu,\text{he}}$ and $\mathbf{u}_{\nu,\text{ta}}$ of size $N_I \times 1$, respectively, and stack $\Gamma[m, l]$ into a matrix $\mathbf{\Gamma}$ of size $\alpha K \times N_I$, and define

$$\mathbf{\Psi} := [\mathbf{\Gamma} \ \mathbf{\Lambda}(T_{\text{bl}}) \mathbf{\Gamma}], \quad \mathbf{u}_\nu[n] := [\mathbf{u}_{\text{he},\nu}^T \ \mathbf{u}_{\text{ta},\nu}^T]^T. \quad (18)$$

Eq. (17) can be compactly expressed as

$$\eta_\nu[n] = \mathbf{\Psi} \mathbf{u}_\nu[n]. \quad (19)$$

Based on the measurements in (15), the least-squares estimate of the interference vector is

$$\hat{\mathbf{u}}_\nu[n] = (\mathbf{\Psi}^H \mathbf{\Psi})^{-1} \mathbf{\Psi}^H \hat{\mathbf{z}}_\nu[n]. \quad (20)$$

Note that $(\mathbf{\Psi}^H \mathbf{\Psi})^{-1} \mathbf{\Psi}^H$ only depends on $\varepsilon_{\max}$, hence can be pre-computed before receiver processing.

2) *Multiuser Channel Estimation and Data Detection*: The desired OFDM component is obtained by subtracting residual interference from the frequency measurements,

$$\begin{aligned} \bar{\mathbf{z}}_\nu[n] &= \mathbf{z}_\nu[n] - \mathbf{\Psi} \hat{\mathbf{u}}_\nu[n], \\ &= \sum_{\mu=1}^{N_u} \mathbf{\Lambda}(\varepsilon_\mu) \mathbf{H}_{\nu,\mu} \mathbf{s}_\mu[n] + \tilde{\mathbf{w}}_\nu[n], \end{aligned} \quad (21)$$

where the equivalent noise term consists of both the ambient noise and the interference estimation error,

$$\tilde{\mathbf{w}}_\nu[n] = \mathbf{\Psi}(\mathbf{u}_\nu[n] - \hat{\mathbf{u}}_\nu[n]) + \mathbf{w}_\nu[n]. \quad (22)$$

For simplicity, we assume that $\tilde{\mathbf{w}}_\nu[n]$ follows a zero-mean complex Gaussian distribution with a covariance matrix $\hat{N}_0[n] \mathbf{I}_{\alpha K}$, where $\mathbf{I}_{\alpha K}$ is an identity matrix of size $\alpha K \times \alpha K$, and $\hat{N}_0[n]$ is the average power of frequency samples at null subcarriers.

With $\{\bar{\mathbf{z}}_\nu[n]\}_{\nu=1}^{N_r}$, the conventional multi-input multi-output (MIMO) OFDM receiver processing is then carried

out. Rather than estimating the channel matrices directly, a sparse channel estimation is achieved by estimating the path parameters specified by the triplet $(A_{\nu,\mu,p}^{(n)}, a_{\nu,\mu,p}^{(n)}, \tau_{\nu,\mu,p}^{(n)})$ in (4) for each receiving hydrophone individually; see, e.g., [10] and [41] for detailed description on the sparse channel estimation. A linear MMSE equalizer [9] and a nonbinary low-density parity-check (LDPC) decoder [42] are used for channel equalization and decoding, respectively. After inputting the LMMSE estimate of information symbols into the nonbinary LDPC decoder, both hard and soft decisions on the information symbols can be obtained, these being fed back for the residual interference estimation, channel estimation and symbol detection in the next iteration. For detailed description on the MIMO-OFDM receiver design, please refer [9] and references therein.

D. Interference Reconstruction

Once the processing of the n th block stops, with the estimated channel matrices and the information symbols the time-domain OFDM waveform in baseband can be reconstructed via the inverse discrete-time Fourier transform,

$$\hat{y}_{\nu,\mu}(t; n) = \sum_{m=-\alpha K/2}^{\alpha K/2-1} \sum_{k=-K/2}^{K/2-1} \hat{H}_{\nu,\mu}[m, k; n] \hat{s}_\mu[k] e^{j2\pi \frac{m}{\alpha T} t}, \quad (23)$$

for $t \in [0, T_{\text{bl}}]$. Based on the aggregated interference representation in (9), estimates of the interference components in $\bar{I}_\nu(t; n-1)$ and $\bar{I}_\nu(t; n+1)$, which are spilled over from the n th block to the $(n-1)$ th and the $(n+1)$ th blocks, respectively, can be obtained by replacing $y_{\nu,\mu}(t; n)$ by $\hat{y}_{\nu,\mu}(t; n)$ in (9). They are then passed to the preceding and the succeeding processing units.

IV. INVESTIGATION ON THE TIME DURATION OF THE AGGREGATED INTERFERENCE

As shown in (9) and (16), the number of degrees of freedom of the interference are decided by its time-bandwidth product $[2B\varepsilon_{\max}]$. In this section, we investigate on the time duration of the interference in an example network.

We consider a network with one data collection unit and multiple sensors, which operates in a *collision-tolerant* fashion by allowing simultaneous transmissions from N_u sensors. For simplicity, we assume that the receiver and sensors are at the same depth in water, the sensors are uniformly distributed within a circle of diameter D_N , and the receiver is located at the origin.

Suppose the network operating according to a MAC protocol with handshaking. The data collection unit first broadcasts the *clear-to-send* (CTS) frame to allow simultaneous transmissions of N_u active sensors which requested to send packets. Once receiving the CTS frame, each sensor starts the data transmission.

Let d_i denote the distance between the i th active sensor and the receiver. Based on the uniform distribution of the i th sensor within the circle, the probability density function (pdf) of d_i will satisfy

$$f(d_i) \propto 2\pi d_i, \quad (24)$$

which leads to the formulation,

$$f(d_i) = \frac{8d_i}{D_N^2}, \quad \text{for } d_i \in [0, D_N/2]. \quad (25)$$

Assuming that the acoustic waveform propagates along a straight line, the time-of-arrival of packets from the i th sensor is thus $\tilde{\varepsilon}_i = 2d_i/c$, with the pdf,

$$g(\tilde{\varepsilon}_i) = \frac{2c^2\tilde{\varepsilon}_i}{D_N^2}, \quad \text{for } \tilde{\varepsilon}_i \in [0, D_N/c] \quad (26)$$

with c being the sound speed in water.

Notice that for the block transmissions, the time-of-arrival of the package $\tilde{\varepsilon}_i$ and the time-of-arrival of each block within the package $\check{\varepsilon}_i$ is related via

$$\check{\varepsilon}_i = [\tilde{\varepsilon}_i]_{\text{mod } T_{\text{bl}}/2} = [2d_i/c]_{\text{mod } T_{\text{bl}}/2}. \quad (27)$$

Take $T_{\text{bl}} = 200$ ms and $c = 1500$ m/s as an example. For $d_i = 15$ m, we have $\check{\varepsilon}_i = 20$ ms; for $d_i = 45$ m, we have $\check{\varepsilon}_i = 60$ ms and for $d_i = 75$ m, we have $\check{\varepsilon}_i = 0$ ms.

Hence, the pdf of $\check{\varepsilon}_i$ is the folded summation of $g(\tilde{\varepsilon}_i)$, with

$$f(\check{\varepsilon}_i) = \sum_{l=0}^L g\left(\check{\varepsilon}_i + \frac{lT_{\text{bl}}}{2}\right) = \sum_{l=0}^L \frac{8(\check{\varepsilon}_i + lT_{\text{bl}}/2)}{D_N^2}, \quad (28)$$

for $\check{\varepsilon}_i \in [0, T_{\text{bl}}/2]$, where $L = \lceil \frac{2D_N}{T_{\text{bl}}c} \rceil$, and the cumulative distribution function (cdf) of $\check{\varepsilon}_i$ follows as

$$F(\check{\varepsilon}_i) = \sum_{l=0}^L \frac{4(\check{\varepsilon}_i^2 + lT_{\text{bl}}\check{\varepsilon}_i)}{D_N^2}, \quad (29)$$

for $\check{\varepsilon}_i \in [0, T_{\text{bl}}/2]$. Substituting (28) and (29) into (31), the distribution of the interference time duration ε_{max} can be obtained.

Assume that the arrival times of N_u users follow an independent and identical distribution with the pdf $f(\check{\varepsilon})$ and cdf $F(\check{\varepsilon})$. The maximum delay ε_{max} is the *range* of the arrival-time sequence [43],

$$\varepsilon_{\text{max}} = \max\{\check{\varepsilon}_1, \check{\varepsilon}_2, \dots, \check{\varepsilon}_{N_u}\} - \min\{\check{\varepsilon}_1, \check{\varepsilon}_2, \dots, \check{\varepsilon}_{N_u}\}, \quad (30)$$

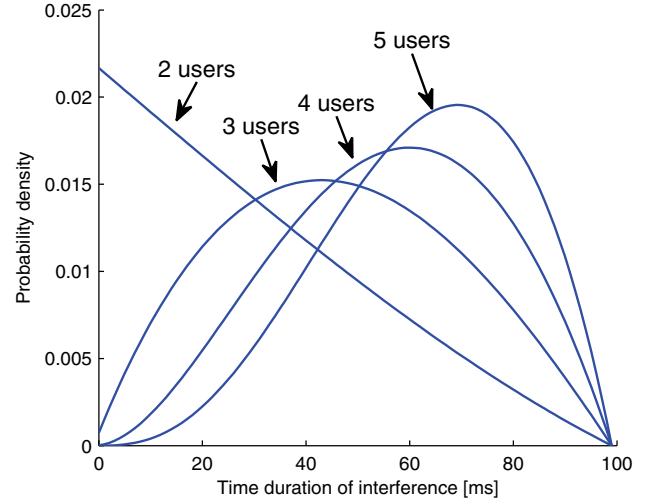
which has the pdf expressed as [43]

$$f_{N_u}(\varepsilon_{\text{max}}) = N_u(N_u - 1) \cdot \int_{-\infty}^{\infty} f(\check{\varepsilon})[F(\check{\varepsilon} + \varepsilon_{\text{max}}) - F(\check{\varepsilon})]^{N_u-2} f(\check{\varepsilon} + \varepsilon_{\text{max}}) d\check{\varepsilon}. \quad (31)$$

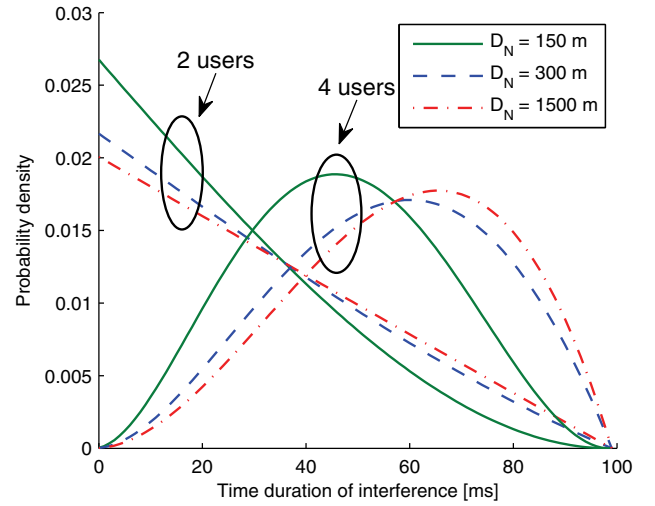
Using the numerical integration, the pdf of ε_{max} corresponding to different number of users is shown in Fig. 4, where D_N takes integer multiples of $T_{\text{bl}}c$. One can see that as the number of users increases, the pdf shifts to the large value region of ε_{max} gradually. A similar trend happens as the diameter D_N increases, but the pdf shifts quite slowly.

V. SIMULATION RESULTS

In simulation, the underwater acoustic channel between each user and each receiving hydrophone during each block transmission is generated randomly. We assume that each channel consists of 10 discrete paths. The inter-arrival-time of paths follows an exponential distribution with a mean of 1 ms. The amplitudes of paths are Rayleigh distributed with the average power decreasing exponentially with the delay,



(a) with different number of users, $D_N = 300$ m



(b) with different values of D_N

Fig. 4. Probability density function of the maximum asynchronism on the OFDM block level in an asynchronous multiuser system, where the users are uniformly distributed within a circle of diameter D_N .

where the difference between the beginning and the end of the guard time is 20 dB. The times-of-arrivals of users' signals follow a uniform distribution within a certain interval. In each simulation instance, we assume that the time-of-arrivals of users' signals are perfectly known at the receiver.

We consider $N_{\text{bl}} = 10$ blocks in each transmission burst. The ZP-OFDM parameters are the following: center frequency $f_c = 13$ kHz, bandwidth $B = 9.77$ kHz, OFDM symbol duration $T = 104.86$ ms, guard interval $T_g = 24.6$ ms, and number of subcarriers $K = 1024$. Out of 1024 subcarriers, there are $|S_N| = 96$ null subcarriers with 24 on each edge of the signal band for band protection and 48 evenly distributed in the middle for the carrier frequency offset estimation; $|S_P| = 256$ are pilot subcarriers uniformly distributed among the 1024 subcarriers, and the remaining are $|S_D| = 672$ data subcarriers for delivering information symbols. The pilot symbols are drawn randomly from a QPSK constellation, and different users have different pilot symbol sets. The

data symbols are encoded with a rate-1/2 nonbinary LDPC code[42] and modulated by a QPSK constellation, which leads to a data rate of each user

$$R = \frac{1}{2} \frac{|\mathcal{S}_D|}{T + T_g} \log_2 4 = 5.2 \text{ kb/user/s.} \quad (32)$$

Throughout this paper, we adopt the block-error-rate (BLER) averaged over all users as the performance metric. A frequency-domain oversampling factor $\alpha = 2$ is used.

The decoding performance of four receiver processing configurations will be compared.

- *Configuration 1*: A block-by-block multiuser reception: By treating the interference as ambient noise, the conventional iterative multiuser decoding techniques in Section III-C2 are used;
- *Configuration 2*: A block-by-block multiuser reception with interference cancellation: By treating the interference as an external interference, the iterative joint multiuser processing and interference cancellation in Section III-C is performed;
- *Configuration 3*: A block-to-block receiver with forward interference cancellation: After the interference subtraction based on the interference estimate from the preceding block, iterative joint multiuser decoding and residual interference cancellation are performed, as in Section III;
- *Configuration 4*: The proposed burst-by-burst receiver with multiple rounds of forward and backward processing.

For fairness of comparison, the frequency-domain oversampling is used for all the configurations, and we set an identical iteration number threshold ($I_{\max} = 4$) in each block processing. Four rounds of forward/backward *block-to-block* processing in configuration 4 are used. In configuration 4, we take 10 blocks within one burst as one batch, while for configurations 1 ~ 3, the batch size can be regarded as one.

In terms of the decoding complexity, one can see that configuration 1 has the lowest complexity, the complexities of configurations 2 and 3 are similar, and configuration 4 has about eight times of the complexity of configuration 3 due to the iterative forward and backward processing. Meanwhile, configurations 1 ~ 3 are capable of on-line processing without decoding latency, while configuration 4 suffers a decoding latency of the burst length.

A. Two-User System with Time-Varying Channels

To explore the receiver performance in the time-varying UWA channels, we draw the Doppler rate of each path independently from a *zero-mean* uniform distribution with standard deviation (std) σ_v m/s according to the path-based channel model in (4). To achieve a good decoding performance, the ICI incurred by the channel variation is considered explicitly. For the sake of receiver complexity, a *band-limitedness* assumption of the channel matrix is adopted by assuming that only elements in the main diagonal and one off-diagonal on each side are nonzero in this setting.

For ICI estimation with regularly distributed pilots, a progressive decoding procedure in [41] is employed. During the iterative processing, the receiver assumes the absence of ICI to

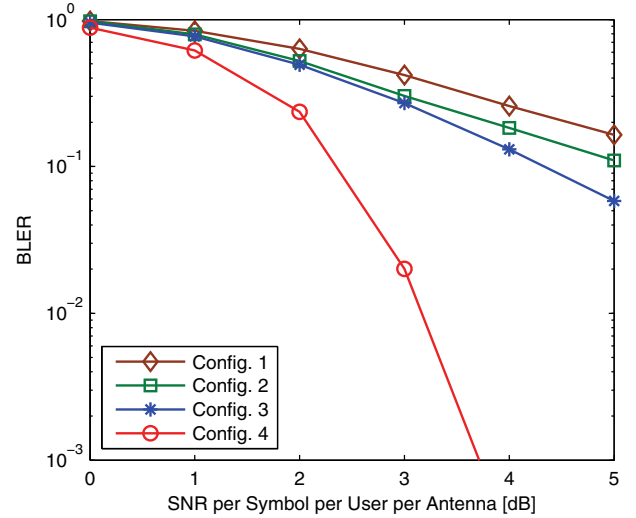


Fig. 5. Block-error-rate performance of four receiving configurations, $\sigma_v = 0.1$ m/s.

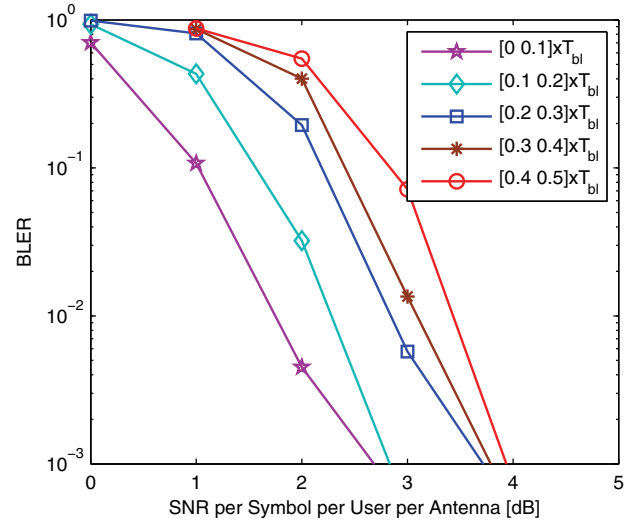


Fig. 6. Block-error-rate performance of the proposed receiver in a two-user system with different relative delays, $\sigma_v = 0.1$ m/s; three receiving hydrophones.

get an initial estimate of the transmitted information symbols. Coupled with pilots, the information symbol estimates are then used in the following iterations for channel estimation.

We assume three receiving hydrophones at the receiver and that the relative delay of the second user is uniformly distributed within the interval $[0, T_{bl}/2]$. In the channel with a mild Doppler spread $\sigma_v = 0.1$ m/s, Fig. 5 shows the BLER performance of the four receiving configurations with different signal-to-noise (SNR) levels. One can see that the conventional receiver without interference cancellation almost cannot work, that the block-by-block interference cancellation brings some performance improvement, and that the burst-by-burst receiver has the best performance. Relative to the one-way message passing in the block-to-block receiver, the two-way message passing in the proposed burst-by-burst receiver improves the decoding performance considerably.

Corresponding to a mild Doppler spread with $\sigma_v = 0.1$ m/s, Fig. 6 shows the BLER performance of the burst-by-burst re-

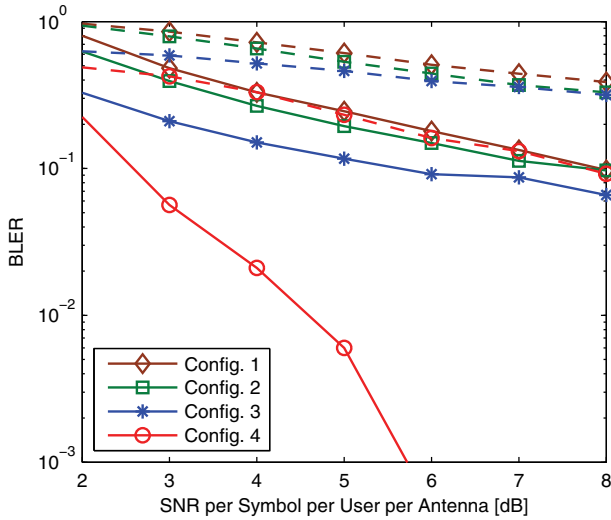


Fig. 7. Block-error-rate performance of four receiving configurations, solid lines: $\sigma_v = 0.3$ m/s; dashed lines: $\sigma_v = 0.5$ m/s.

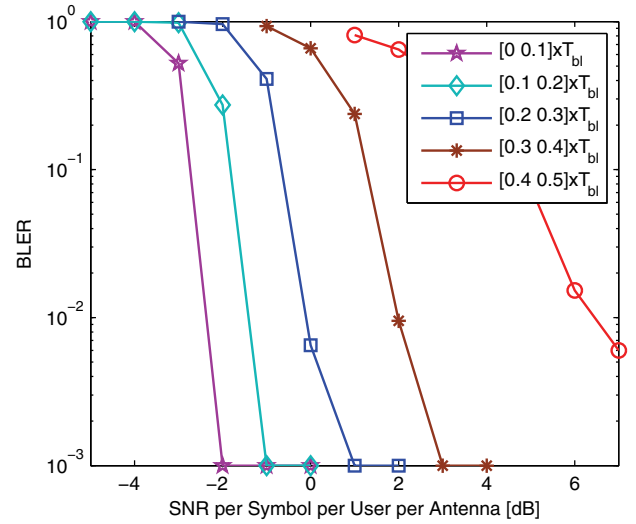


Fig. 9. Block-error-rate performance of the proposed receiver in a four-user system with different relative delays, six receiving hydrophones.

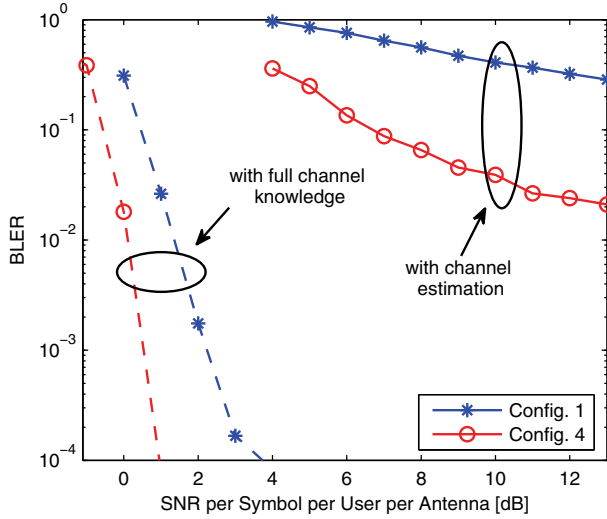


Fig. 8. Block-error-rate performance of two receiving configurations with and without perfect channel knowledge, two receiving hydrophones are used, and $\sigma_v = 0.3$ m/s.

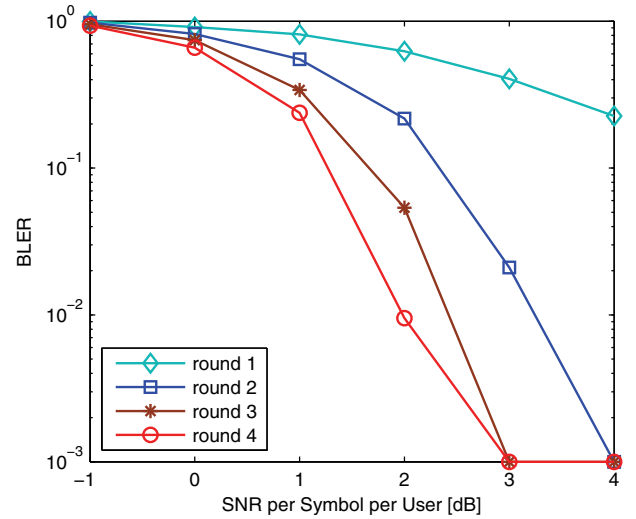


Fig. 10. Block-error-rate performance of the proposed receiver with four rounds of forward/backward processing, $\varepsilon_{\max} \sim \mathcal{U}[0.3, 0.4] \times T_{bl}$, four users, six receiving hydrophones.

ceiver when the relative delay of the second user is uniformly distributed within five consecutive intervals: $[0, 0.1]T_{bl}$, $[0.1, 0.2]T_{bl}$, $[0.2, 0.3]T_{bl}$, $[0.3, 0.4]T_{bl}$ and $[0.4, 0.5]T_{bl}$. One can see that as the relative delay of the second user, i.e., the time duration of the interference increases, the required SNR for successful decoding of the two data streams also increases.

With three receiving hydrophones, the BLER performance of the four receiving configurations in the channel with severe Doppler spreads $\sigma_v = 0.3$ m/s and $\sigma_v = 0.5$ m/s are shown in Fig. 7. Similar to the observation in the channel with a mild Doppler spread, the proposed burst-by-burst receiver with forward and backward message passing outperforms other configurations considerably.

Assuming two receiving hydrophones, the BLER performance of the conventional receiver and the proposed receiver with and without perfect channel knowledge is shown in

Fig. 8. Relative to the scenario with channel estimation, one can see that the performance gap between the two receiving schemes decreases when perfect channel knowledge is available. One could infer that channel estimation accuracy degrades drastically if the interblock interference is not accounted for.

B. Multi-User System with Time-Invariant Channels

To examine the performance of the proposed burst-by-burst receiver as a function of number of users, we set $\varepsilon_1 = 0$, and assume that the relative delay of the second user is uniformly distributed within a certain interval, and that the delays of users $3 \sim N_u$ are uniformly distributed between zero and the upper bound of this interval. By dividing half of the OFDM block duration $[0, T_{bl}/2]$ into five intervals: $[0, 0.1]T_{bl}$, $[0.1, 0.2]T_{bl}$, $[0.2, 0.3]T_{bl}$, $[0.3, 0.4]T_{bl}$ and $[0.4, 0.5]T_{bl}$, Figs. 9 and 10 show the BLER performance of the proposed

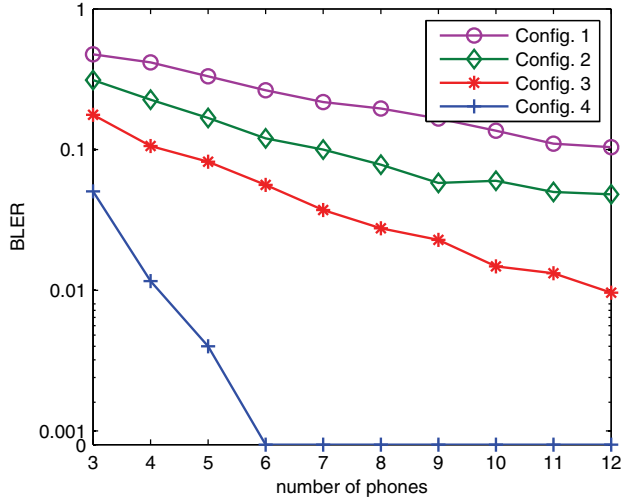


Fig. 11. Block-error-rate performance of four receiving configurations, MACE10 data sets.

receiver with four asynchronous users and six hydrophones at the receiver. Relative to the BLER performance in the two-user scenario in Fig. 6, one can see that as the number of users increases, the maximum delay of the users, i.e., the time duration of the interference has more impact on the decoding performance. Meanwhile, one can also observe a considerable performance improvement brought by the iterative forward and backward message passing.

VI. EMULATED EXPERIMENTAL RESULTS

For the receiver performance evaluation, we use the data set collected in a point-to-point experimental setting to emulate the data set collected in an asynchronous multiuser communication system. The *mobile acoustic communication experiment* (MACE10) was carried out off the coast of Martha's Vineyard, Massachusetts, June, 2010. The water depth was about 80 meters. The receiving array was stationary, while the source was towed slowly away from the receiver and then towed back, at a speed around 1 m/s. The relative distance of the transmitter and the receiver changed from 500 m to 4.5 km [41]. Out of the two tows in this experiment, we only consider the data collected in the first tow. There are 31 transmissions in total, with 20 blocks in each transmission. We exclude one transmission file recorded during the turn of the source, where the SNR of the received signal is quite low. The average SNR of considered files is around 20 dB.

The ZP-OFDM parameters are the following: center frequency $f_c = 13$ kHz, bandwidth $B = 4.883$ kHz, OFDM symbol duration $T = 209.7$ ms, guard interval $T_g = 40.3$ ms, and number of subcarriers $K = 1024$. The subcarrier distribution is identical to that in the simulation setting. Different blocks have different pilot symbol sets. With a rate-1/2 nonbinary LDPC code and a QPSK constellation for information bit encoding and mapping, the data rate of each user is

$$R = \frac{1}{2} \frac{|\mathcal{S}_D|}{T + T_g} \log_2 4 = 2.7 \text{ kb/user/s.} \quad (33)$$

To test the proposed receiver for asynchronous N_u -user transmissions, we generate a *semi-experimental* data set by

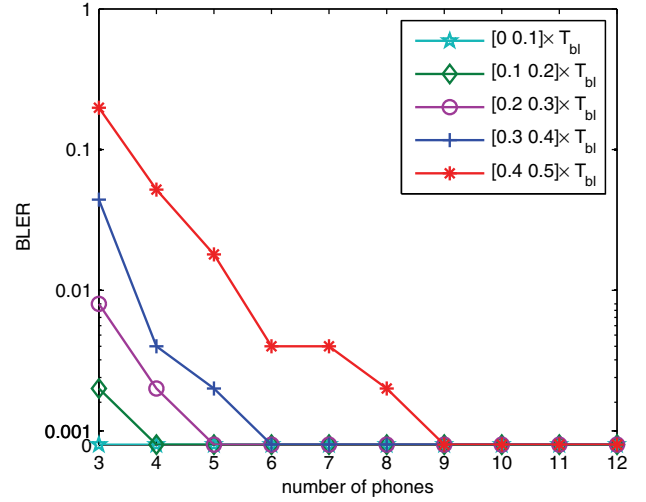


Fig. 12. Block-error-rate performance of the proposed receiver with different relative delays, four rounds of forward and backward processing and eight iterations within each block processing; MACE10 data sets.

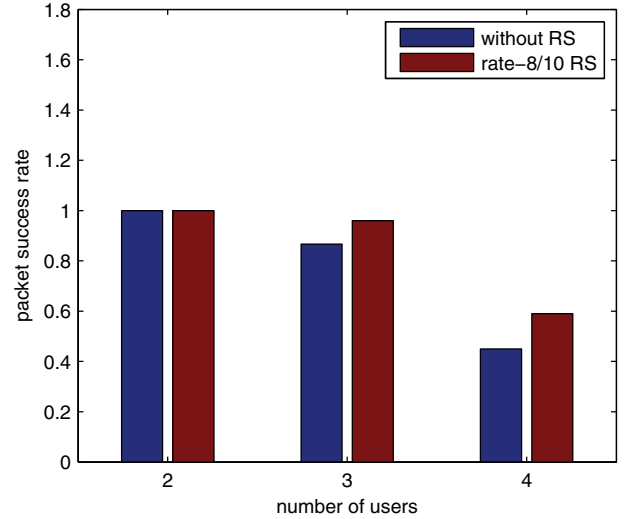


Fig. 13. Packet-success-rate of the proposed receiver with different number of users with and without a rate-8/10 Reed-Solomon erasure-correction code across 10 blocks, four rounds of forward/backward processing, eight iterations within each block processing.

consecutively dividing received data blocks within each transmission into N_u groups and regarding the blocks within each group as the signal from each user. The received semi-experimental waveform is formed by adding the N_u groups together. A resampling operation is then used to remove the Doppler effect caused by the mobility of the source array. Note that due to the existence of ambient noise in the received data blocks, the SNR per user in the semi-experimental data set decreases according to $10 \log(N_u)$ dB as the number of users increases.

Similar to the simulation setup, the relative delay of each user is uniformly distributed within a certain interval. By setting the distribution interval corresponding to the second user as $[0, T_{bl}/2]$, Fig. 11 shows the decoding performance of the four configurations in the simulation, and four rounds of forward/backward *block-to-block* processing in configuration 4

are used. Again, one can see that the conventional multiuser reception approach without interference cancellation almost cannot work, the block-by-block interference cancellation method improves the performance slightly, and the proposed burst-by-burst receiver with the forward and backward message passing is the best.

With different distribution intervals of the relative delay, the BLER performance of the proposed burst-by-burst receiver with two asynchronous users is shown in Fig. 12. One can see that as the relative delay increases, i.e., the time duration of the interference increases, more receiving hydrophones are required for successful decoding.

Assuming that the relative delays of users are uniformly distributed within $[0, T_{bl}/2]$, Fig. 13 shows the packet-success-rate of the proposed receiver with different number of users. One can see that as the number of users increases, the decoding performance gets worse gradually. The degradation can be attributed to the increased multiuser interference and the increased ambient noise power due to the generation of the semi-experimental data sets.

To achieve a robust decoding performance of the proposed receiver, a block-level Reed-Solomon (RS) erasure-correction code over Galois field can be used as an inter-block code while the nonbinary LDPC code is used as an intra-block code [44]; the optimal combination between the erasure- and error-correction codes for the layered coding approach has been studied in [45]. With a rate-8/10 block-level *shortened* RS codeword³ applied for each data subcarrier across a packet consisting of 10 blocks, any two blocks can be received in error (hence erased), while the whole data burst can be recovered. Fig. 13 shows the packet-success-rate (PSR) of the proposed receiver with different number of users. Compared with the packet-success-rate without using erasure-correction coding, introducing two redundant blocks leads to a considerable performance improvement. This option is appealing for practical systems with asynchronous users.

VII. CONCLUSIONS

In this paper, we proposed an asynchronous multiuser reception method for OFDM in underwater acoustic communications. With an overlapped truncation of the received signal and using the *interference aggregation* concept, the asynchronous multiuser problem was converted to a synchronous multiuser problem with interference, where interference mitigation and multiuser decoding are carried out in an iterative fashion. Extensive simulation and emulated experimental results demonstrated that the proposed receiver can effectively recover the transmitted symbols from multiple asynchronous users. The decoding performance can be further improved by using a block-level RS erasure-correction code.

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³A shortened RS code can be obtained by setting some information symbols as zeros from a RS code of longer length [44].

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